

Econ 310 Extra Credit Assignment

Introduction: Saving money is an important thing to do especially facing emergency consequences such as being unemployed. During the pandemic, many people were forced to quit their jobs. In this period of time, people might be afraid that they are not able to pay rent or buy daily necessities. Under a high unemployment rate, will people save their money more than usual? The purpose of this research is to examine the relationship between the unemployment rate (UNRATE) and personal saving rate (PSAVERT).

a. Main Hypothesis: As the unemployment rate increases, the personal saving rate is expected to rise. The relationship between unemployment rate and personal saving rate is positive.

b. Describe the relevant population:

The unemployment rate data (UNRATE) from the FRED website focuses on people in the United States who are 16 years or older. This group includes everyone who is either working or actively looking for a job. However, It does not include people who are not working and not looking for a job, such as retirees and full-time students, etc.

The Personal Saving Rate (PSAVERT) shows the percentage of money that people in the United States save from their income after paying taxes. It looks at how much money is left after people pay for their daily expenses and bills. It may be viewed as the portion of personal income that is used either to provide funds to capital markets or to invest in real assets such as residences.

Data Description: The data for this project is sourced from publicly available U.S. economic datasets:

1. **Unemployment Rate (UNRATE):** Data from FRED.

Federal Reserve Bank of St. Louis. "Unemployment Rate." *FRED, Federal Reserve Bank of St. Louis*, <https://fred.stlouisfed.org/series/UNRATE>.

2. Personal Saving Rate (PSAVERT): Data from FRED.

Federal Reserve Bank of St. Louis. "Personal Saving Rate." *FRED, Federal Reserve Bank of St. Louis*, <https://fred.stlouisfed.org/series/PSAVERT>.

Both data are collected on FRED website, I combined the three dataset together by using the spreadsheet tool. I cleaned up the dataset by sorting the two dataset with the same time period and making sure that there is no data missing or misplaced. The final sample consists of monthly observations over the period 1959/01/01 to 2024/09/01. In the final data, it contains 789 observations and 4 variables.

3. Descriptive Statistics:

According to Table 1 in the Appendix section, it shows that the dataset contains 789 observations and 3 variables, which are DATE (date), PSAVERT (personal saving rate), and UNRATE (unemployment rate).

4. Analysis

a. Write out your hypothesis that you will test

Null Hypothesis (H_0): There are no relationship between the unemployment rate (UNRATE) and the personal saving rate (PSAVERT). $\beta_1 = 0$

Alternative Hypothesis (H_1): There is a relationship between the unemployment rate (UNRATE) and the personal saving rate (PSAVERT). $\beta_1 \neq 0$

b. Describe in detail which statistical analysis you have selected and why it is appropriate.

I used the regression analysis and supported my result by using the correlation analysis to show the relationship between unemployment rate and personal saving rate. The regression analysis shows how one variable changes when the other variable changes. If the p-value is less than 0.05, which is the significant value, then there is a relationship between the two variables because when one variable increases, the other variable also increases. If the p-value is greater than 0.05, then it indicates that there is no relationship between the two variables. I also double checked the result with the correlation analysis, which also tells the relationship between the two variables. When the correlation coefficient is positive, then the relationship between the two variables is positive and when the correlation coefficient is negative, then the relationship between the two variables is negative.

c. Conduct the appropriate analysis

The STATA code for regression analysis is: `reg unrate psavert` (See Table 2)

The STATA code for correlation analysis is: `cor unrate psavert` (See Table 3)

Show results & Conclusion

According to table 2 in the Appendix, the p-value is 0.000. This shows that the correlation between unemployment rate and saving rate is positive and significant because it is less than 0.05. According to table 3 in the Appendix, the correlation test is supporting the same result that we get in the regression analysis. In table 3, the correlation is 0.2337, which is greater than 0, indicating a positive relationship between the two variables. This shows that the two variables are positively correlated. Because of this, we have enough evidence to reject the null hypothesis

and support the alternative hypothesis. Therefore, there is a positive relationship between the unemployment rate (UNRATE) and the personal saving rate (PSAVERT).

6. Works Cited

Federal Reserve Bank of St. Louis. "Unemployment Rate." *FRED, Federal Reserve Bank of St. Louis*, <https://fred.stlouisfed.org/series/UNRATE>..

Federal Reserve Bank of St. Louis. "Personal Saving Rate." *FRED, Federal Reserve Bank of St. Louis*, <https://fred.stlouisfed.org/series/PSAVERT>.

7. Appendix

Table 1) Describe the dataset

```
. describe
```

Contains data

Observations: **789**

Variables: **3**

Variable name	Storage type	Display format	Value label	Variable label
date	str10	%10s		DATE
psavert	float	%9.0g		PSAVERT
unrate	float	%9.0g		UNRATE

Sorted by:

Note: Dataset has changed since last saved.

Table 2) Regression Analysis

```
. reg unrate psavert
```

Source	SS	df	MS	Number of obs	=	789
Model	179.543551	1	179.543551	F(1, 787)	=	68.62
Residual	2059.04972	787	2.61632747	Prob > F	=	0.0000
				R-squared	=	0.0802
				Adj R-squared	=	0.0790
Total	2238.59327	788	2.8408544	Root MSE	=	1.6175

unrate	Coefficient	Std. err.	t	P> t	[95% conf. interval]
psavert	.1387383	.0167478	8.28	0.000	.1058627 .1716139
_cons	4.716139	.1526573	30.89	0.000	4.416476 5.015803

Table 3) Correlation Analysis

```
. cor unrate psavert if psavert < 20
(obs=786)
```

	unrate	psavert
unrate	1.0000	
psavert	0.2337	1.0000

Table 4) Data

	date	psavert	unrate
1	1959-01-01	11.3	6
2	1959-02-01	10.6	5.9
3	1959-03-01	10.3	5.6
4	1959-04-01	11.2	5.2
5	1959-05-01	10.6	5.1
6	1959-06-01	10.5	5
7	1959-07-01	10.7	5.1
8	1959-08-01	9.6	5.2
9	1959-09-01	8.7	5.5
10	1959-10-01	9.4	5.7
11	1959-11-01	10.1	5.8
12	1959-12-01	11	5.3
13	1960-01-01	10.9	5.2
14	1960-02-01	10.6	4.8
15	1960-03-01	9.4	5.4
16	1960-04-01	8.4	5.2
17	1960-05-01	10.4	5.1
18	1960-06-01	10.4	5.4
19	1960-07-01	10.4	5.5
20	1960-08-01	10.3	5.6
21	1960-09-01	10	5.5
22	1960-10-01	9.8	6.1
23	1960-11-01	9.8	6.1
24	1960-12-01	10.4	6.6
25	1961-01-01	11.1	6.6
26	1961-02-01	11.1	6.9
27	1961-03-01	10.4	6.9